ATHARVA PRASHANT KALE

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SUMMARY

Embedded Systems and Firmware Engineer specializing in hardware-software integration, automated test script development (Python, C), and low-voltage PCBA validation. Proven track record of debugging complex communication peripherals (CAN, Ethernet, I2C) using oscilloscopes and logic analyzers. Adept at translating electrical schematics into safety-critical logic for high-vibration automotive and industrial environments.

EDUCATION

Northeastern University, Boston, MA **Dec 2026**
Candidate for Master of Science in Internet of Things
Courses: Wireless Sensor Networks & IoT, Computer Engineering Fundamentals, Computer Hardware Security, Machine Learning/Pattern Recognition

MES Pillai College of Engineering, Mumbai, India **May 2023**
Bachelor of Engineering in Electronics and Telecommunications Engineering

TECHNICAL SKILLS

- **Programming Languages & Scripting:** Python, C, C++, Embedded C, Verilog, VHDL, Assembly, MATLAB, Bash Scripting
- **Microcontrollers & FPGA:** MSP430, ESP32, ARM Cortex-M, Raspberry Pi, Tang Nano 20K (GOWIN), Xilinx FPGAs, LPC2148
- **Embedded Protocols & RTOS:** Embedded Linux, FreeRTOS, CAN, I2C, SPI, UART, RS-485, GPIO, MQTT, TCP/IP, Ethernet
- **Test & Validation Tools:** Oscilloscope, Logic Analyzer, DAQ, Hardware-in-the-Loop (HIL), Multi-meter
- **Hardware Design:** Altium Designer, SolidWorks PCB, Git, Vivado, GOWIN IDE

RELEVANT EXPERIENCE

Massachusetts Department of Transportation (MassDOT) – Boston, MA **Jan 2026 - Present**
Highway Engineering Co-op – District 6 (M/E/C Building Systems)

- **Prevented hardware rejection** by identifying proprietary SFP transceiver validation risks and specifying compliant single-mode fiber standards.
- **Validated physical-to-digital signal** interpretation by mapping 4-20mA analog outputs from infrared sensors to automated facility control logic.
- **Resolved physical hardware constraints** by evaluating integration tradeoffs between native PLC software ecosystems and 4-port fiber architectures.
- **Ensured 100% facility safety compliance** prior to Lower Explosive Limit (LEL) thresholds by executing field-level validation protocols and replacing failed infrared sensors across the District 6 automated gas detection network.

Broad Net India Pvt. Ltd. – Mumbai, India **Jan - Jun 2023**
Hardware and Network Engineering Intern

- **Analyzed** Linux kernel logs and TCP/IP packet loss metrics to resolve edge device routing failures, optimizing throughput for commercial clients.
- **Conducted** signal attenuation testing on fiber-optic links using optical power meters, ensuring physical medium integrity met ISP standards.
- **Authored** technical SOPs for signal testing procedures, reducing team onboarding time and standardizing diagnostic protocols.

ACADEMIC PROJECTS

FPGA-Based Ring Oscillator PUF for Secure Key Generation **Jan - Apr 2025**
Technologies: Verilog, Tang Nano 20K (GW2A-LV18QN88C8I7), GOWIN IDE, MATLAB, Oscilloscope

- Designed and synthesized a hardware security primitive (PUF) in **Verilog**, utilizing 128 ring oscillator pairs to generate device-unique cryptographic keys from silicon manufacturing variations.
- Optimized logic placement constraints to achieve **85% entropy and 95% reliability**, validating bit stability against voltage/temperature fluctuations using MATLAB analysis.
- Resolved synthesis timing violations and verified clock frequency behavior using hardware-in-the-loop testing on the Tang Nano 20K development board.
- Developed MATLAB scripts to automate test response collection and evaluate entropy, streamlining the hardware validation and bit-stability analysis process.

Formula SAE - Hyperion Racing Team (SAE) **May 2022 - Feb 2023**
Technologies: Proteus, EagleCAD, SolidWorks PCB, SolidWorks Electrical, DAQ, Oscilloscope

- **Validated** PCBA signal integrity by designing a fail-safe Brake System Plausibility Device and executing Hardware-in-the-Loop (HIL) testing to simulate high-vibration automotive conditions.
- **Analyzed** complex electrical schematics to develop the complete low-voltage automotive wiring harness, ensuring signal reliability under high-vibration track conditions.
- **Debugged** in-track electrical issues during endurance testing, coordinating sensor calibration fixes with the powertrain team.

LEADERSHIP ACTIVITIES

Electronics Head of Hyperion Racing Team, Formula SAE Team **Jun 2022 - Apr 2023**